

Thabang L. Mashinini-Sekgoto

LEAD DATA SCIENTIST · APPLIED AI, DATA SCIENCE & AI ENGINEERING

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Summary

Lead Data Scientist with 9+ years turning data, machine learning and optimisation into systems that run in production and deliver measurable value, across banking, telecommunications, applied research and higher education.

What I have done. Architected the data engineering capability that data science at Absa runs on, with Ubunye Engine processing 100M+ events a day for 10+ products. Built Vodacom's Smart Generator Optimisation engine across 15,000+ sites, contributing about R1 billion a year. Published climate machine learning research at NeurIPS from IBM Research.

What I offer. One person who can set the strategy, build the platform, grow the team and ship the product, from first idea to governed production. MSc in Computer Science (Distinction); PhD in Computer Science from 2027.

Education

University of the Witwatersrand

Johannesburg, South Africa

PHD IN COMPUTER SCIENCE (COMMENCING 2027)

2027

- Proposed research: physics informed self supervised learning for SAR based flood extent mapping, for climate risk in data scarce regions.

University of the Witwatersrand

Johannesburg, South Africa

MSC IN COMPUTER SCIENCE (DISTINCTION)

2018 – 2019

- Thesis: 📄 [Learning Level Set Method by Echo State Network for Image Segmentation](#). A spatiotemporal deep learning formulation of level set segmentation, benchmarked across five architectures and four datasets.

University of the Witwatersrand

Johannesburg, South Africa

BSc HONOURS IN COMPUTER SCIENCE

2017

- Project: Wildfire Estimation Using Kernel Density Estimators.

University of the Witwatersrand

Johannesburg, South Africa

BSc IN COMPUTATIONAL & APPLIED MATHEMATICS AND ASTRONOMY

2014 – 2016

Experience

Absa Group

Johannesburg, South Africa

LEAD DATA SCIENTIST

Mar. 2024 – Present

- Took data science from infrastructure not built for big data, with no documentation or shared practice, to a modern platform: led the modernisation strategy and **architected the data engineering capability from scratch**, bringing Databricks into the bank with CI/CD and integration with legacy systems, as part of the move from on premises to cloud (AWS and Databricks).
- That capability runs on **Ubunye Engine**, which processes **100M+ events a day** and is the engineering layer for **10+ products**.
- Modernised telematics, about **7 to 10 million trip records a day**, cutting processing from **months to under a day**; modernised the flood forecasting used by underwriting on the same platform.
- Leading the hyper personalisation strategy for clients, building customer lifetime value and segmentation models.
- Lead a team of **3** data scientists; introduced AI and model governance aligned to the bank's standards, and set the best practices, documentation and training the team works from.

Vodacom

Johannesburg, South Africa

SENIOR DATA SCIENTIST

Nov. 2021 – Mar. 2024

- Led and trained a team of **10+** graduate data scientists in network operations, on optimisation, network downtime and network quality.
- Built the **Smart Generator Optimisation** engine (📄 [product](#)) from scratch: mathematical optimisation and graph traversal to prioritise and rank **15,000+ sites** nationally, contributing about **R1 billion a year** in savings. Recognised with the **Vodacom Star Award**.
- Built a multi output traffic forecasting model (Gaussian mixture) predicting voice and data traffic across 2G, 3G and 4G for all **15,000+ sites**, used for revenue allocation: attributing to each tower the revenue it generates.
- Built the real time framework behind it: Kafka feeds from generators and towers enriched in flight with enterprise data lake sources, using PyFlink and PySpark.
- Built **Smart Planning**, an optimisation engine that simulates demand and supply under constraints to decide where to build new towers, on a computational graph of the whole network and its dependencies.

IBM Research

Johannesburg, South Africa

MACHINE LEARNING RESEARCH SCIENTIST

Apr. 2020 – Nov. 2021

- Machine learning for climate forecasting, built to beat the current climate model baselines; published at 📄 [NeurIPS 2020](#) and 📄 [EGU 2022](#).
- Built towards 📄 [IBM PAIRS](#), processing petabytes of satellite imagery and other data on IBM high performance computing clusters.
- Contributed to the Gauteng COVID-19 risk index and prediction dashboard (IBM Research Africa, Wits, GCRO), used by the Gauteng Department of Health to find hotspots and plan hospital capacity.
- Worked in an international research team with scientists from across the globe.

University of the Witwatersrand, Business Intelligence Services

Johannesburg, South Africa

DATA SCIENTIST

Jun. 2018 – Apr. 2020

- Built a course recommendation engine that matched students to courses from their matric results, used by student advisers and linked to about **R2 million a year** in government subsidy.
- Built student success models that helped heads of school plan budgets and enrolment to meet subsidy requirements.
- Trained graduates in the vacation work programme and institutional researchers from across the country.

Council for Scientific and Industrial Research (CSIR)

Pretoria, South Africa

DATA SCIENTIST & SOFTWARE ENGINEER

Nov. 2017 – Jan. 2018

- Built dashboards and AI analytics on municipal finance, youth unemployment and municipal inefficiency with researchers from several universities, used by **17** municipalities including the City of Cape Town.
- Recognised for innovation by the [Mail & Guardian](#), the [CSIR](#) and the [Department of Science and Technology](#).

ZAR X

Johannesburg, South Africa

SYSTEMS ANALYST & SUPPORT

Mar. 2017 – Nov. 2017

- Systems analysis and operational support for a newly launched stock exchange serving 13,000+ clients, through go live.

Skills

Leadership	Data and AI strategy, building capabilities from scratch, team leadership and training, AI and model governance, stakeholder management
Machine Learning	Forecasting, mathematical optimisation, graph algorithms, recommender systems, segmentation and lifetime value, anomaly detection, Bayesian filtering, geospatial and climate modelling
Data Engineering	Apache Spark and PySpark, Databricks, Delta Lake, Kafka, Apache Flink and PyFlink, high performance computing
MLOps & Platform	MLflow, Unity Catalog, Databricks Asset Bundles, CI/CD with GitHub Actions, Docker, Kubernetes, AWS
Generative AI	LLM applications (Claude, Gemini, OpenAI), retrieval augmented generation, AI agents
Languages	Python, SQL, TypeScript, R, C/C++

Publications

Long-Range Seasonal Forecasting of 2m-Temperature with Machine Learning

NEURIPS 2020, TACKLING CLIMATE CHANGE WITH ML WORKSHOP

2020

- Machine learning for long range seasonal temperature forecasting, for climate intelligence. [Paper](#)

ML-based Probabilistic Prediction of 2m Temperature and Total Precipitation

EGU GENERAL ASSEMBLY 2022

2022

- Probabilistic machine learning forecasts of temperature and rainfall. [Abstract](#)

Mine Worker Threshold Shift Estimation via Optimization Algorithms for Deep Recurrent Neural Networks

IFAC MMM 2019

2019

- Optimisation and recurrent neural networks to estimate hearing loss risk for mine workers. [Paper](#)

Products & Open Source

Ubunye AI Ecosystems: Ubunye Engine

PRINCIPAL STAFF RESEARCH ENGINEER

2024 – Present

- Open source, config driven data and ML framework: one transform class and a YAML file, and the same pipeline runs on a laptop, Docker, Kubernetes, cloud clusters or Databricks. Every run records a receipt (schema hash and data fingerprint) so runs can be compared.
- Runs in production at the bank and is the engineering layer for most platforms I build; takes pipelines from prototype to deployment in under 8 hours instead of about 2 weeks. 700+ tests, published on PyPI.
- [Documentation](#) [Source](#) [PyPI](#)

tfilterspy

CREATOR & PRINCIPAL ENGINEER

2023 – Present

- Built after years of working with noisy sensor and telemetry data: Kalman, Extended, Unscented, Ensemble and Particle filters behind one scikit-learn style interface, learning noise settings from data, with parallel filters on Dask for high volume streams.
- [Documentation](#) [Source](#) [PyPI](#)

Teaching & Public Engagement

FabAcademic Unfiltered, with Prof. Mamokgethi Phakeng

INVITED AI EDUCATOR

2024 – Present

- Teach the practical use of AI and AI agents in everyday life to an audience of **300,000+** globally, hands on, with live demos, slides, code, tutorials and open discussion; 17 recorded talks. [🔗 YouTube](#) [🔗 Talks](#)
- Built **LeparaLaMapara**, an AI assistant grounded in my public work, and write on AI and data engineering for a general audience. [🔗 Assistant Writing](#)

Honors & Awards

2026	Judge, AI Champion Award , Make Education Fashionable (MEF) Convocation	MEF
2022	Vodacom Star Award , For the Smart Generator Optimisation engine	Vodacom
2019	MSc with Distinction , Outstanding academic and research performance	University of the Witwatersrand
2018	Recognised for innovation , Mail & Guardian, CSIR and the Department of Science and Technology	CSIR
2014 – 2017	Co-Founder, Wits Astronomy Club , STEM outreach and public astronomy engagement	Wits University

References

Dr.	Etienne Vos (IBM Research Manager) , 🔗 Etienne.vos@ibm.com	IBM Research
Dr.	Jacobus Du Toit (Machine Learning Lead) , 🔗 Jaco.Dutoit@vodacom.co.za	Vodacom
Dr.	Akram Zaytar (Senior Research Scientist) , 🔗 akramzaytar@microsoft.com	IBM/Microsoft Research
Mrs.	Harshila Dulabh (Team Lead) , 🔗 Harshila.Dulabh@wits.ac.za	Wits University (BIS)